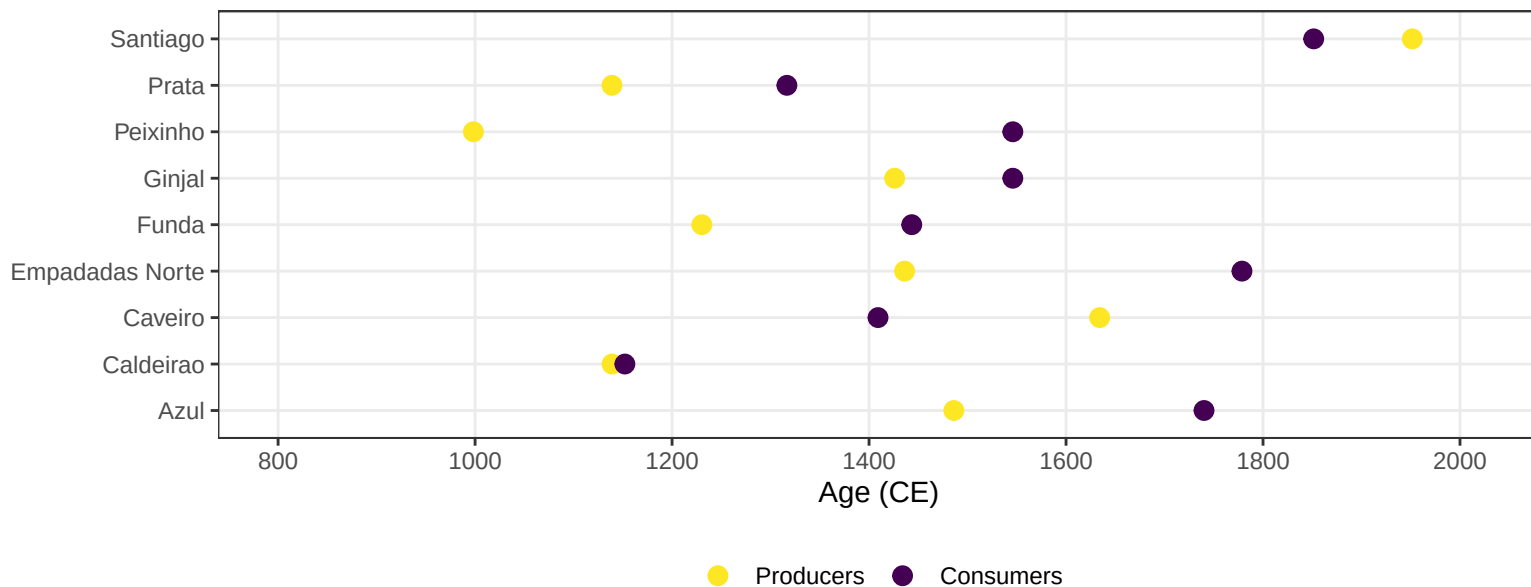


(a)**First significant derivative onset (≥ 800 CE)**

From per-lake DCA1 GAMs (same records as Fig. 2b); earlier change intervals excluded by design

**(b)****Onset difference (producer – consumer, ≥ 800 CE)**Bar colour = earlier group; grey = similar timing ($|\text{diff}| \leq 25$ yr)